

Computer Vision:

Week-1 (12-16 Jan) Theory Lectures

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Basics of Computer Vision (CV)

- This course is crucial due to its wide range of **applications in AI**, such as self-driving cars, medical imaging, and retail automation. It provides foundational knowledge for building systems that interpret and make decisions from visual data (images, videos).
- The field is in high demand across industries like healthcare, security, and manufacturing. CV is essential for roles like **AI engineer** and **machine learning researcher**, with growth in automation and AI innovation.

Basics of Computer Vision (CV)

- It also helps in developing strong programming and problem-solving skills while learning important libraries such as OpenCV and PyTorch.
- Computer Vision fosters cross-disciplinary knowledge, working with areas like **robotics and neuroscience**.
- It is a field of constant **advancement**, providing opportunities for research and innovation. Mastery of CV ensures a future-proof career in an AI-driven world. Ultimately, CV is a key technology powering smart systems, from **object recognition** to real-time video analysis.)

Introduction to Computer Vision and Its Applications:

- Computer Vision (CV) is a branch of Artificial Intelligence that enables machines to perceive, analyze, and understand visual information from images and videos.
- It aims to replicate human visual perception using computational models.
- CV combines concepts from image processing, machine learning, linear algebra, and optimization.

Goals of Computer Vision:

- Extract meaningful information from visual data
- Recognize and classify objects and patterns
- Understand scenes and activities
- Support decision-making in intelligent systems.

Difference Between Image Processing and Computer Vision:

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Difference Between Image Processing and Computer Vision:

Image Processing	Computer Vision
Enhances images	Understands image content
Low-level operations	High-level interpretation
Example: noise removal	Example: object recognition

Applications of Computer Vision:

- *Medical Imaging:* tumor detection, X-ray and MRI analysis
- *Autonomous Vehicles:* lane detection, obstacle avoidance
- *Surveillance Systems:* face recognition, motion tracking
- *Industrial Automation:* defect detection, quality control
- *Agriculture:* crop monitoring, disease detection
- *Smart Cities:* traffic monitoring, crowd analysis.

Role of CV in Artificial Intelligence:

- Acts as the eyes of AI systems
- Integrates with Deep Learning (CNNs, Vision Transformers)
- Enables intelligent perception in real-world environments.

Human Visual System, Image Formation, and Image Representation:

- Human Visual System (HVS)
- The human eye captures light reflected from objects
- Retina contains rods (light sensitivity) and cones (color perception)
- Brain interprets visual signals via the visual cortex.

Comparison of Human Vision and Computer Vision:

Human Vision	Computer Vision
Adaptive and robust	Depends on algorithms
Perception-based	Computation-based
Handles noise naturally	Needs preprocessing

Image Formation Process:

- Light source illuminates an object
- Object reflects or absorbs light
- Camera sensor captures reflected light
- Image is formed as a 2D projection of a 3D scene.

$$Image = Illumination \times Reflectance$$

Camera and Image Acquisition:

- Lens focuses light onto the sensor
- Sensor converts light into electrical signals
- Signals are digitized to form images
- Common sensors: CCD, CMOS.

Image Representation:

- Digital image is a 2D array of pixels
- Each pixel stores intensity or color information
- Pixel values represent brightness or color levels.

Vision System Pipeline and Challenges:

- *Image Acquisition*
- *Pre-Processing*
- *Feature Extraction.*
- *Feature Representation.*
- *Learning / Inference.*
- *Post-processing & Output*

Image Acquisition

- Source: Camera, CCTV, satellite, medical scanners
- Considerations:
 - Resolution
 - Frame rate
 - Sensor type (CCD / CMOS)
 - Lighting conditions

Pre-processing

-Common operations:

- Noise removal (Gaussian, Median filtering)
- Contrast enhancement (Histogram Equalization)
- Normalization
- Image resizing and color space conversion (RGB $\rightarrow$ Gray, HSV)

Feature Extraction

Extract meaningful information from images.

-Types of features:

- **Low-level:** Edges, corners, textures
- **Mid-level:** Shapes, regions, keypoints
- **High-level:** Objects, faces, actions

Feature Representation

Transform raw features into a compact form.

-Methods:

- Feature vectors
- Histograms (HOG)
- Embeddings (deep learning)

**Good representation improves classification accuracy*

Learning/Inference

Decision-making stage.

-Methods:

- Classical:
 - k-NN
 - SVM
 - Decision Trees
- Deep Learning:
 - CNNs
 - Vision Transformers (ViT)

Image Formation & Camera Models

Contents

- Image Formation
- Pinhole Camera Model
- Perspective Projection
- Coordinate Systems

Why do we study camera models?

The real world is 3D

Images are always 2D

A camera converts 3D scene $\rightarrow$ 2D image

Computer Vision must understand this conversion

What is Image Formation?

Image is formed because:

Light hits objects

Objects reflect light rays

Camera collects those rays

Sensor converts rays into pixels

Why Not Just Open a Camera Without Hole/Lens?

If the camera opening is large:

Many rays from different points mix

Image becomes blurred

Pinhole Camera Model?

Think of a box with a tiny hole:

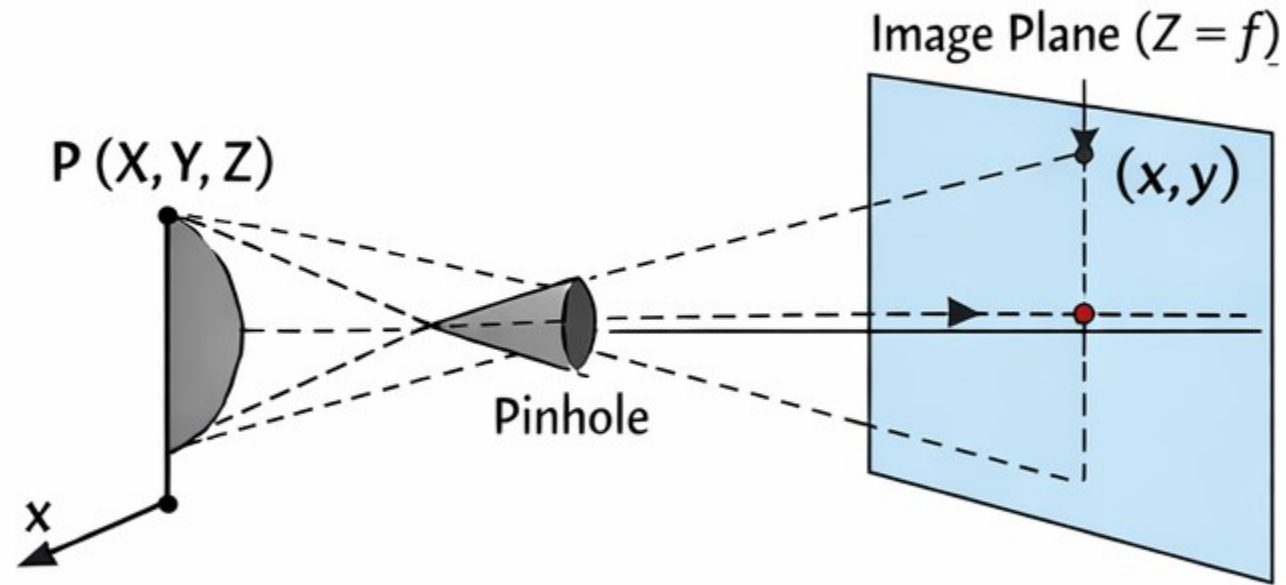
Light passes through the hole

Rays meet at one point (pinhole)

Clear image forms on image plane

Pinhole Camera

Pinhole Camera Model



Pinhole Camera: 3D to 2D Mapping

How Does It Convert $3D \rightarrow 2D$?

A 3D point:

Extracted Point Data

3D Space Point: $P(X, Y, Z)$

2D Image Plane Point: (x, y)

Pinhole Camera: 3D to 2D Mapping

The Projection Relationship

In a standard pinhole camera model, the relationship between the 3D coordinates and the 2D coordinates on the image plane is typically defined by the following equations (assuming a focal length f)

Pinhole Camera: 3D to 2D Mapping

$$x = f \frac{X}{Z}$$

$$y = f \frac{Y}{Z}$$

Perspective Projection

Perspective means:

Far objects look smaller

Near objects look bigger

Example:

A building far away looks tiny

Same building from close looks huge

This is natural in all real cameras.

Coordinate Systems

In Computer Vision, we use different coordinate frames:

World Coordinates (X_w, Y_w, Z_w)

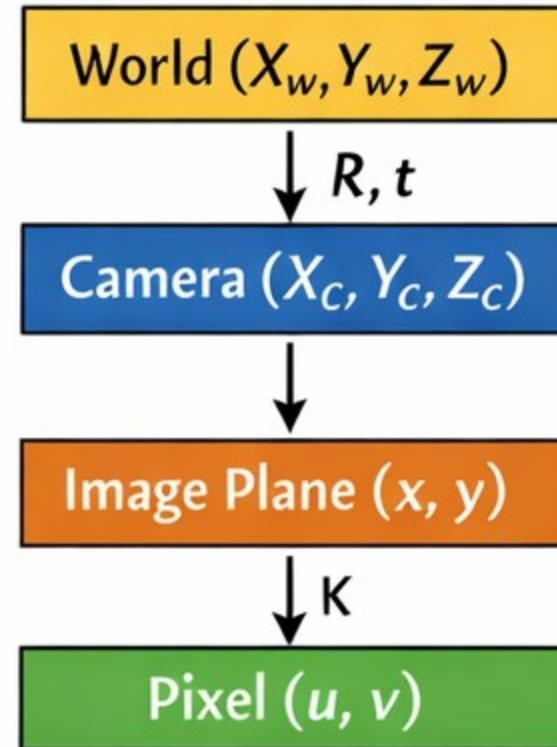
Camera Coordinates (X_c, Y_c, Z_c)

Image Plane (x, y)

Pixel Coordinates (u, v)

Coordinate System

Coordinate Systems



Coordinate Systems

In Computer Vision, we use different coordinate frames:

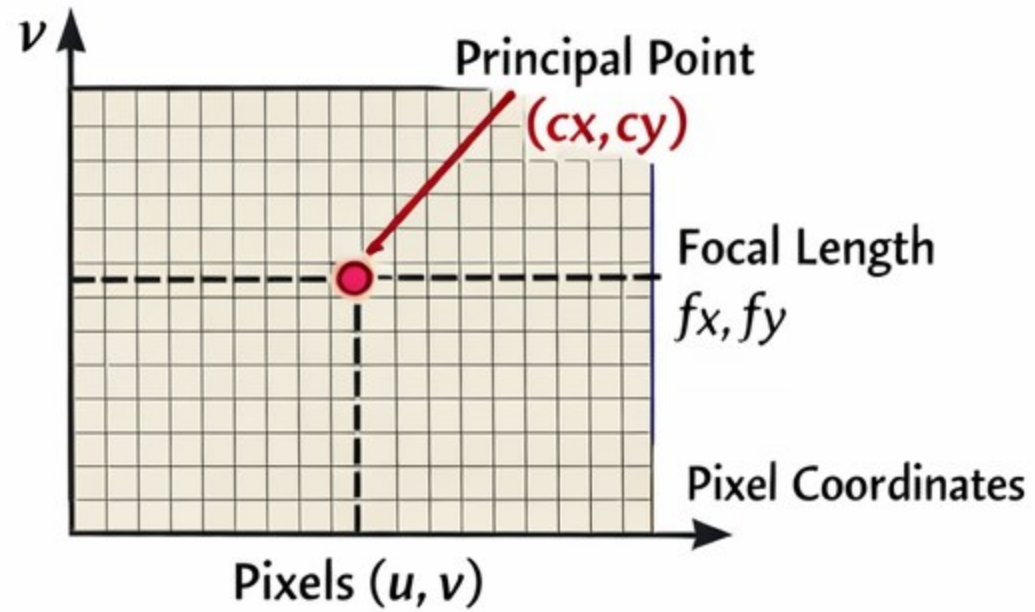
Coordinate System	Definition
World Coordinates	The real-world position relative to a fixed origin in the physical environment.
Camera Coordinates	The position as seen from the camera, where the camera center is the origin (0,0,0).
Image Plane	A continuous 2D point on the physical sensor plane (usually measured in millimeters).
Pixel Coordinates	The final point inside a digital image, represented by discrete pixel locations.

Camera intrinsics

Camera intrinsics describe how a camera converts a 3D real-world scene into a 2D image.

Camera intrinsics

Camera Intrinsics (K)



Camera intrinsics

Suppose

- Your **eye** is the camera
- The **retina** is the image sensor
- The **lens** focuses light

Camera intrinsics tell us:

- How strong the lens is (zoom)
- Where the center of the image is
- How pixels are shaped

Camera intrinsics

*The image shows a standard
camera intrinsic matrix, typically denoted as K .*

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$